

Comparative analysis of Biportal Endoscopic L5-S1 foraminotomy vs Biportal endoscopic L5-S1 fusion

Sun's orthopedics
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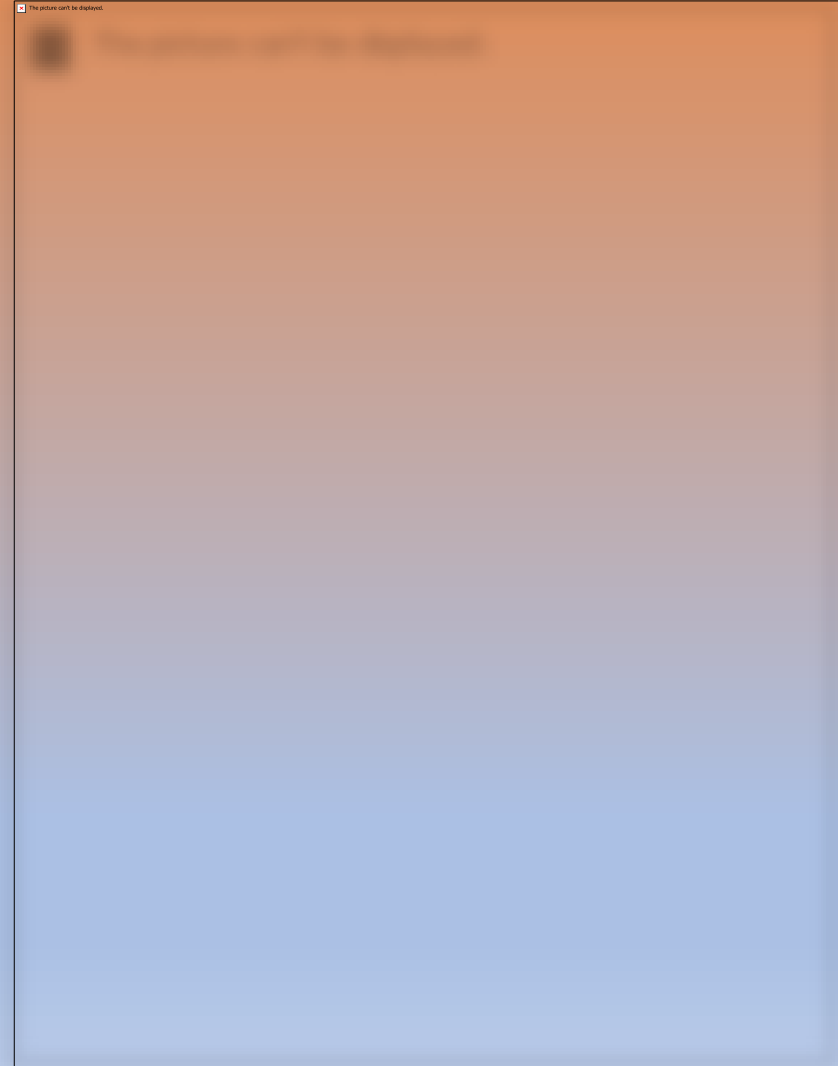
Introduction and purpose

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- Common problem in low back : L5-S1
- Unique anatomy of L5S1
- Pro for L5_S1 foraminotomy
- Sx free longevity after L5-S1 foraminotomy
- L5-S1 foraminotomy vs L5-S1 fusion
- My Experience

Materials and methods

- Materials
 - L5-S1 paraspinal foraminotomy group
 - May. 2017 - Apr. 2022
 - 18 patients
 - Age Average 69.2(48-87)
 - Mean F/U 5.6 yrs
 - L5-S1 fusion group
 - Jan. 2021- Aug. 2023
 - 19 patients
 - Age Average 64.6(51-82)
 - Mean F/U 2.3 yrs
- Methods
 - Pre and post op clinical scores
 - VAS back, VAS leg, ODI, McNab criterial
 - T-test, ManWhitney U test



Foraminotomy Case F/48



- Back pain
- Lt thigh pain
- Difficult to walk
 - DTR (+/+)
 - SLR 70/70
 - Sensory WNL/WNL
 - MOTOR BTDF(5/5)
BTPF(5/5)

Pre op

Post op

Post OP Sx

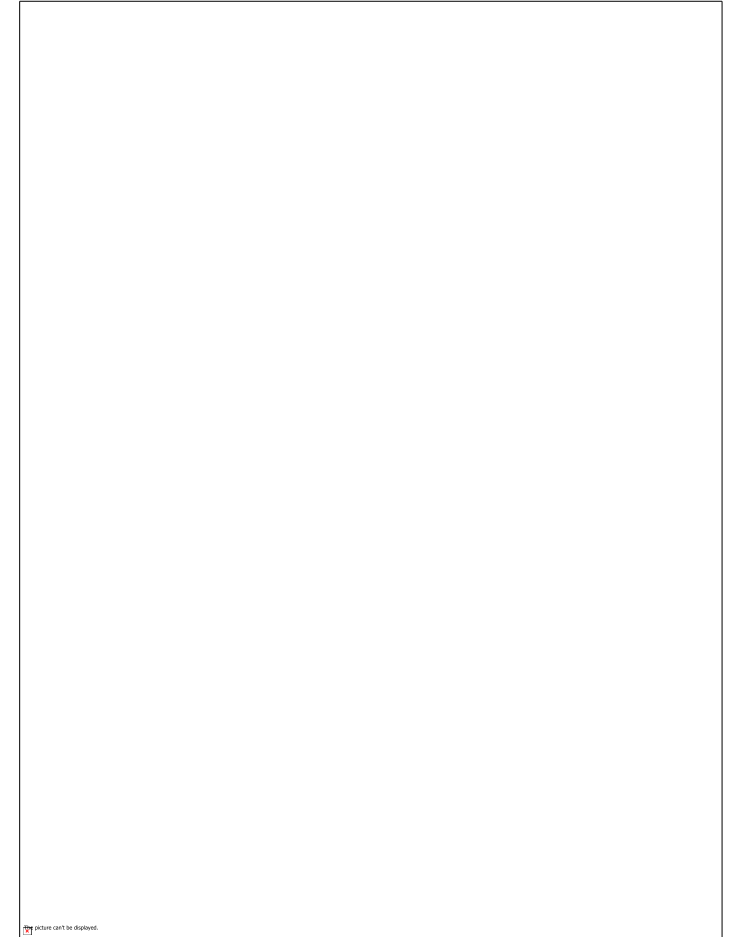
- PO 1 month : improved back, buttock, thigh pain
- PO 2month: back pain, limping, numbness
- PO 4 yrs : painful limping after 10 minutes walking

Fusion case pre op sx M/53

- Sx develop , 5 yrs ago
- both thigh posterior pain, numbness, tightness, heaviness
- No back pain, both buttock pain
- Lt sx more severe
- Lt Ankle dorsiflexion weakness (4)
 - DTR (++/+++)
 - SLR 70/70
 - Sensory WNL/WNL
 - MOTOR BTDF(5/4)
BTPF(5/5)

Facet sacrificing TLIF approach

- **Anterior position of cage**
- **Transverse lie**
- **Pedicle to pedicle position in AP view**



Width
17 mm



Length
45 mm

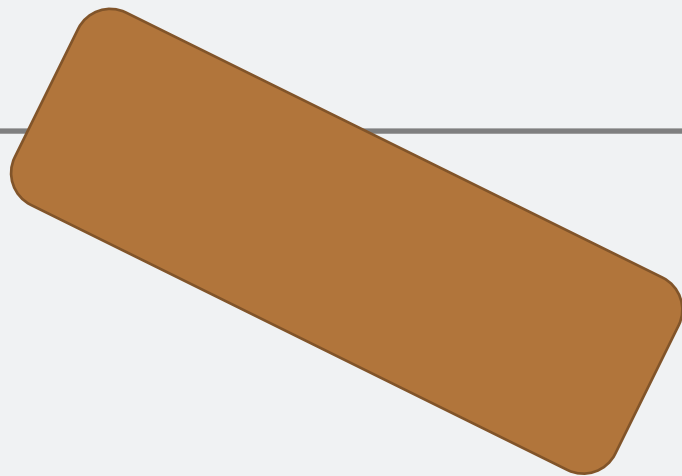


Height 13 mm

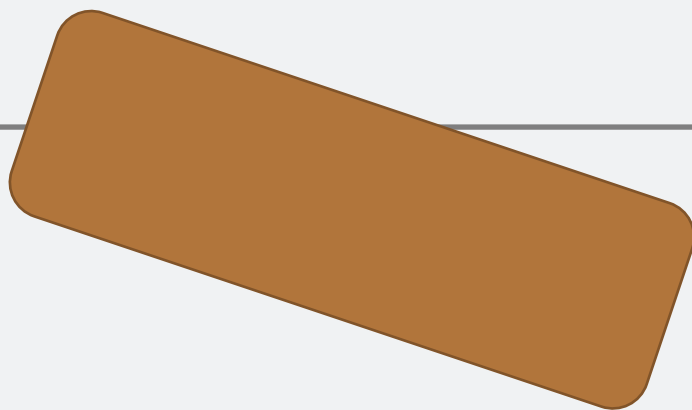
11 mm

9 mm

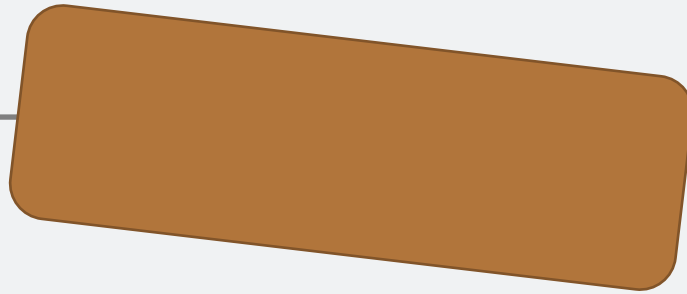
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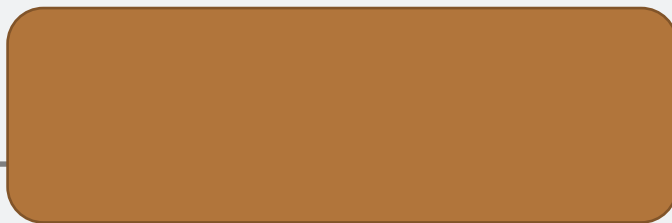
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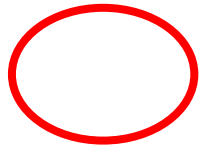


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Pre op

Post op

Post op sx - po 3m , 1 yr

- Slight numbness under the knee, posterior leg
- Ankle dorsiflexion improved
- No buttock pain

Results

Limitations

- No double blind prospective study
- Operation period - difference
- Technique difference
- Multilevel case in both groups
- Different F/U period

Conclusion

- L5-S1 foraminotomy
 - Early sx recurrence
 - 2nd operation
- L5-S1 fusion
 - Early outcome : good
 - Junctional problem
 - Cost
 - Operation time : longer than foraminotomy group